

Quantum Annealing for Multiobjective Query Optimization

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Abstract

Given a query plan, the goal of multiobjective query optimization [1] is to optimally map each operator in a plan to an implementation. Operator implementations differ by quality metrics such as cost and latency. In this context, query optimization becomes a multiobjective optimization problem. The optimal solution depends on user preferences and minimizes, for instance, a weighted sum over different cost metrics. We show how to transform instances of multiobjective query optimization, an NP-hard problem [1], into equivalent Quadratic Unconstrained Binary Optimization (QUBO) formulas that can be solved efficiently on quantum annealers.

ACM Reference Format:

Immanuel Trummer. 2026. Quantum Annealing for Multiobjective Query Optimization. In *The third workshop on Quantum Computing and Quantum-Inspired Technology for Data-Intensive Systems and Applications (Q-Data '26)*, May 31–June 05, 2026, Bengaluru, India. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3811628.3811837>

Background. Quantum annealers generate solutions to QUBO problems quickly. A QUBO instance is defined by a quadratic formula that depends on binary variables. The goal is to assign variables to values such that the formula value is minimized. To solve optimization problems via quantum annealing, they must be mapped to QUBO formulas such that an optimal QUBO solution represents an optimal solution to the original problem.

Multiobjective Query Optimization. Query plans are represented as binary trees. Leaf nodes represent data retrieval operations, whereas inner nodes represent join operators. The cost and latency are directly given for leaf nodes. For inner (join) nodes, the accumulated cost is the sum of the costs of both sub-trees and the selected join operator. The latency is the sum of the maximum of the latencies of the two sub-trees and the latency of the selected join operator. We assume that cost and latency values are represented as integers. The goal of multiobjective query optimization is to minimize the weighted sum $\alpha \cdot L + \beta \cdot C$, where L is latency and C is cost, and $\alpha, \beta \in [0, 1]$ are user-defined weights.

QUBO Variables. For each possible implementation j for operator i , we introduce a binary QUBO variable O_{ij} that is set to one iff the implementation is selected for the operator. Also, for each operator i , we introduce variables L_{ik} such that the sum $\sum_k L_{ik} \cdot 2^k$ represents the latency of executing the sub-tree rooted at operator i . Finally, for each join operator i , we introduce “slack variables” S_{1ik}

and S_{2ik} , used to represent idle time spent waiting for the second join operand after the first operand has already been produced.

QUBO Terms Representing Constraints. Problem constraints must be encoded as terms in the QUBO formula that take high values if constraints are violated. First, the QUBO formula must contain terms taking high values if operators are not assigned to implementations. The quadratic term $(1 - \sum_j O_{ij})^2$, to be introduced for each Operator i , reaches a minimal value of zero iff each operator is assigned to exactly one implementation. For a join operator i (i.e., an inner node in the query plan tree), denote by λ_1 and λ_2 the latency for the left and the right sub-tree, respectively (λ_1 and λ_2 correspond to constants for leaf nodes and to a weighted sum $\sum_k 2^k \cdot L_{i'k}$ for join operator i' as operand). If selecting a join operator with latency λ_3 to implement Operator i , the latency for the corresponding sub-tree is given as $\max(\lambda_1, \lambda_2) + \lambda_3$. We must introduce QUBO terms to ensure that $\sum_k 2^k \cdot L_{ik}$ is assigned to that value. The term

$$\left(\sum_k 2^k \cdot L_{ik} - \lambda_1 - \lambda_3 - \sum_k 2^k \cdot S_{1ik} \right)^2 \quad (1)$$

can take its minimal value of zero iff $\sum_k 2^k \cdot L_{ik} \geq \lambda_1 + \lambda_3$. If the time required for producing the second join operand is higher than for the first (i.e., $\lambda_2 > \lambda_1$), it is $\sum_k 2^k \cdot L_{ik} > \lambda_1 + \lambda_3$. However, Eq. 1 can reach a value of zero for suitable settings of the slack variables S_{1ik} . We introduce the equivalent term to Eq. 1 for the second join operand $((\sum_k 2^k \cdot L_{ik} - \lambda_2 - \lambda_3 - \sum_k 2^k \cdot S_{2ik})^2)$. Together, both terms motivate QUBO solutions that choose values for variables L_{ik} such that $\sum_k 2^k \cdot L_{ik}$ at least matches the latency $\max(\lambda_1, \lambda_2) + \lambda_3$.

QUBO Terms Representing Costs. For latency, we add the term $\alpha \cdot \sum_k 2^k \cdot L_{ik}$ where i represents the index of the operator at the root of the tree. It weighs the latency of the entire plan tree by the user-defined weight α . For cost, we add the term $\beta \cdot \sum_{ij} C_{ij} \cdot O_{ij}$, summing over all operators and implementations, where C_{ij} represents the cost of Implementation j for Operator i and β is the user-defined weight for cost.

Prioritizing QUBO Terms. Denote by T_C the terms representing constraints and by T_O the terms representing the objective function. The final QUBO formula has the form $W_C \cdot T_C + T_O$, associating constraint terms with a sufficiently high weight to prioritize them in the optimal QUBO solution. Whenever constraints are violated, T_C must take a value of at least one. Hence, choosing W_C sufficiently high ensures that no difference in the objective function can make up for constraint violations. Choosing W_C higher than the sum of all cost and latency values satisfies that constraint.

References

- [1] CH Papadimitriou and M Yannakakis. 2001. Multiobjective query optimization. In *PODS*. 52–59. <https://doi.org/10.1145/375551.375560>



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ACM ISBN 979-8-4007-2703-0/26/05

<https://doi.org/10.1145/3811628.3811837>